

Fractal Geometry: 0/0 at Every Point on the Boundary

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The Removable Singularity Framework

Abstract

We show that the Mandelbrot set is the universal object connecting all 0/0 removable singularities. Its boundary has Hausdorff dimension $D=2$ (space-filling!), meaning it has 0/0 singularities at EVERY POINT. The Mandelbrot set $z \rightarrow z^2 + c$ exhibits self-similarity: zoom into the boundary and find the same structure at all scales. Julia sets transition from connected (inside) to Cantor dust (outside) at the boundary -- a 0/0. Benoit Mandelbrot discovered BOTH fractal geometry (the Mandelbrot set) AND fractal finance (fat tails) -- the same fractal thinking applies to both. This connects to turbulence (fractal dimension), SOC (fractal avalanches), E8 (critical points), and zeta zeros (critical line).

1. The Mandelbrot Set

$$z_{\{n+1\}} = z_n^2 + c$$

The Mandelbrot set is the set of complex numbers c for which the iteration $z \rightarrow z^2 + c$ remains bounded. Inside: bounded (stable). Outside: diverges (unstable). On the boundary: 0/0 removable singularity. The boundary has Hausdorff dimension $D=2$ (space-filling!) and length infinity. It is connected and self-similar.

2. Escape Time

$$N(c) = \min\{n : |z_n| > 2\}$$

The escape time $N(c)$ measures how quickly a point escapes to infinity. Inside the Mandelbrot set: $N = \text{infinity}$ (never escapes). Outside: $N = \text{finite}$ (escapes quickly). On the boundary: 0/0 (borderline). The escape time creates the famous colored images of the Mandelbrot set.

3. Julia Sets

For each complex number c , the Julia set $J(c)$ is defined by the same iteration. If c is inside the Mandelbrot set: $J(c)$ is connected. If c is outside: $J(c)$ is Cantor dust (totally disconnected). At the boundary: 0/0 (transition between connected and disconnected). The Mandelbrot set is the "parameter space" of all Julia sets.

4. Fractal Dimensions

- Line: $D = 1$ (smooth)
- Koch snowflake: $D = \log(4)/\log(3) \sim 1.26$
- Sierpinski triangle: $D = \log(3)/\log(2) \sim 1.59$
- Mandelbrot boundary: $D = 2$ (SPACE-FILLING!)

The Mandelbrot boundary has the maximum possible dimension for a curve in the plane. It is space-filling, meaning it has 0/0 at EVERY POINT. This is the most exotic 0/0 in our framework.

5. Self-Similarity

Zoom into the Mandelbrot boundary: the same structure appears at all scales. Mini-Mandelbrot sets are embedded everywhere. This is the hallmark of fractals: self-similarity across scales. The 0/0 singularity repeats at every magnification.

6. Same Mandelbrot: Geometry and Finance

Benoit Mandelbrot discovered BOTH fractal geometry (the Mandelbrot set, 1980) AND fractal finance (fat tails in cotton prices, 1963). The same fractal thinking applies to both:

- Mandelbrot set: self-similar, 0/0 everywhere
- Fractal finance: self-similar volatility, 0/0 at crashes
- Both: power laws, scale invariance, removable singularities

7. Connections to Prior 0/0 Singularities

Fractal	Prior Chapter	Connection
Mandelbrot finance	Ch.38 Finance	Same Mandelbrot!
Fractal dimension	Ch.37 Turbulence	D of flow
Fractal avalanches	Ch.41 SOC	Power laws
Critical points	Ch.24 E8	Lie algebras
Critical line	Ch.25 Zeta	$\text{Re}(s)=1/2$
Fractal clusters	Ch.36 Ising	Cluster D

8. Conclusion

The Mandelbrot set is the universal object connecting all 0/0 removable singularities. Its boundary has Hausdorff dimension $D=2$ (space-filling), meaning it has 0/0 at EVERY POINT. Self-similarity means the 0/0 repeats at all scales. Benoit Mandelbrot discovered both fractal geometry and fractal finance -- the same fractal thinking applies to both. The Mandelbrot set connects to turbulence (fractal dimension), SOC (fractal avalanches), E8 (critical points), and zeta zeros (critical line). It is the most beautiful and universal 0/0 in all of mathematics.

References

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